NPR1: a hypothesis on its 3D structural model and a preliminary classification of NPR1 protein sequences across different species

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INTRODUCTION

NPR1 is the key factor of Salicylic acid (SA) signalling, induced in plants under abiotic and biotic stress conditions. Infection by pathogen and subsequent increasing in SA concentration trigger NPR1 activation, whose mechanism of action is showed in Figure 1.

NPR1 by *A. thaliana* binds SA with high *in vitro* affinity (K_d=140 nm), perhaps through Cys 521-529 residues (Wu et al., 2012).

Here we present the modeling of 3D structure of NPR1, the identification of hypothetical binding pockets with SA and the assessment of conservation degree of functional residues between NPR1 proteins of different species.

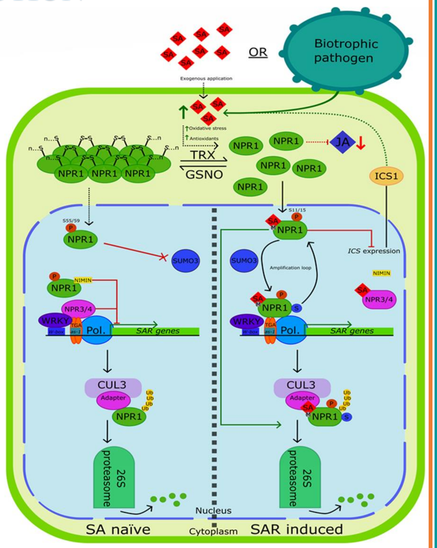

Figure 1: NPR1 activation and its mechanism of action (adapted from Backer et al., 2019)

METHODS

The work was divided into the following main steps:

1) preliminary structural and functional analysis of NPR1 protein;

2) structural modeling, visualization and energy/stereochemistry validation;

3) rigid docking simulations;

4) study of functional residues conservation between different species.

1
- GenBank
- UniProt
- BioGRID
- IntAct
- STRING
- InterPro
- Pfam
- PANTHER
- PROSITE
- PSIPRED4.0
- PDB
- Blast
- ClustalOmega
- T-COFFEE

2
- Modeller10.1
- Phyre2
- AlphaFold
- Prosa-web
- QMEANDisCo
- PROCHECK
- PyMol

3
- PubChem
- Autodock 4.2.5.1
- Autodock tool 1.5.6
- AutoGrid 4.2.5.1
- OpenBabelGUI

4
- Encyclopedia of Life
- MEGA-6
- Meme-suite

Figure 2: Workflow with relative tools applied in this project

RESULTS

1 Preliminary analysis of NPR1 protein

The preliminary analysis allowed to identify four domains, the main functional residues and the secondary structure (Figure 3) of NPR1 protein by *A. thaliana*.

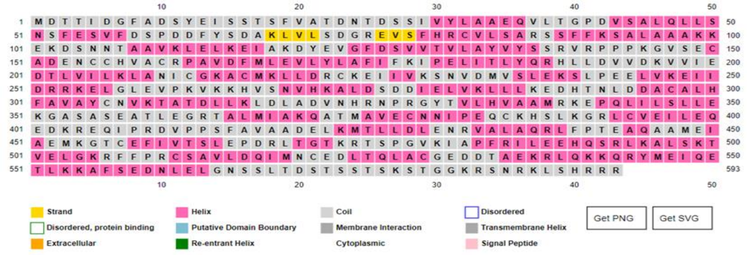

Figure 3: Secondary structure prediction of NPR1 protein using PSIPRED 4.0

2 Structural modeling

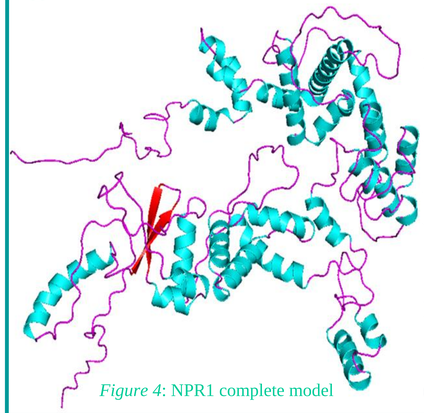

Figure 4: NPR1 complete model

Each functional domain has been modeled using comparative modeling or fold-recognition strategy. The best models obtained have been integrated with NPR1 structure predicted by AlphaFold to extract a 3D model for whole NPR1 protein. The resulting model recorded a z-score of -6.89, a QMean Disco global score of 0.39±0.05, value linked to destructured loops, and only 0.4% of residues in not allowed regions.

3 Rigid docking simulations

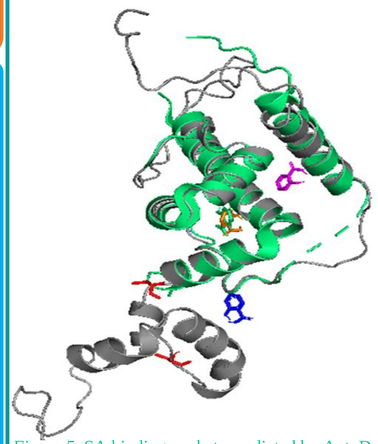

Figure 5: SA binding pockets predicted by AutoDock

In the rigid body docking simulation focused on NPR1/NIM1-like domain, three hypothetical binding pockets between NPR1 and SA were highlighted.

The first conformation recorded better energy and population values, while the second one, despite the higher energy, is allocated in proximity of Cys521-529. The third pocket, instead, has been reported because SA has the same position recorded in the known structure of NPR4 protein crystallized with SA.

4 Study of functional residues conservation between different species

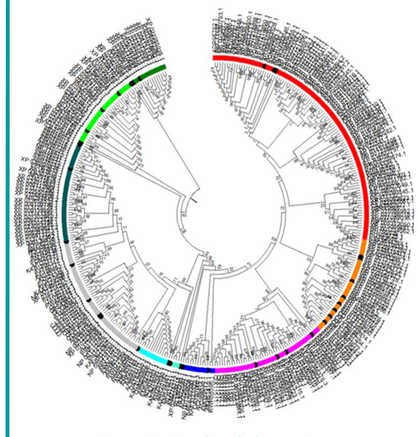

Figure 6: Circular phylogenetic tree

A clustering analysis among 354 protein sequences has been carried out to value the degree of conservation between different species. Ten main clades, each one with characteristic functional motifs and logos, have been identified. This analysis showed that NPR1 protein has a poor conservation of functional residues. Surely, it would be proper to deepen this study by expanding the number of involved proteins.

REFERENCES

- Backer R., Naidoo S. and Van den Berg N., *The NONEXPRESSOR OF PATHOGENESIS-RELATED GENES 1 (NPR1) and Related Family: Mechanistic Insights in Plant Disease Resistance*, Frontiers in Plant Science, 2019, (DOI: 10.3389/fpls.2019.00102)
- Wu Y., Zhang D., Yan Chu J., Boyle P., Wang Y., Brindle I. D., De Luca V. and Despre´ C. *The Arabidopsis NPR1 protein is a receptor for the plant defense hormone salicylic acid*, Cell Reports, 2012, (DOI: 10.1016/j.celrep.2012.05.008)